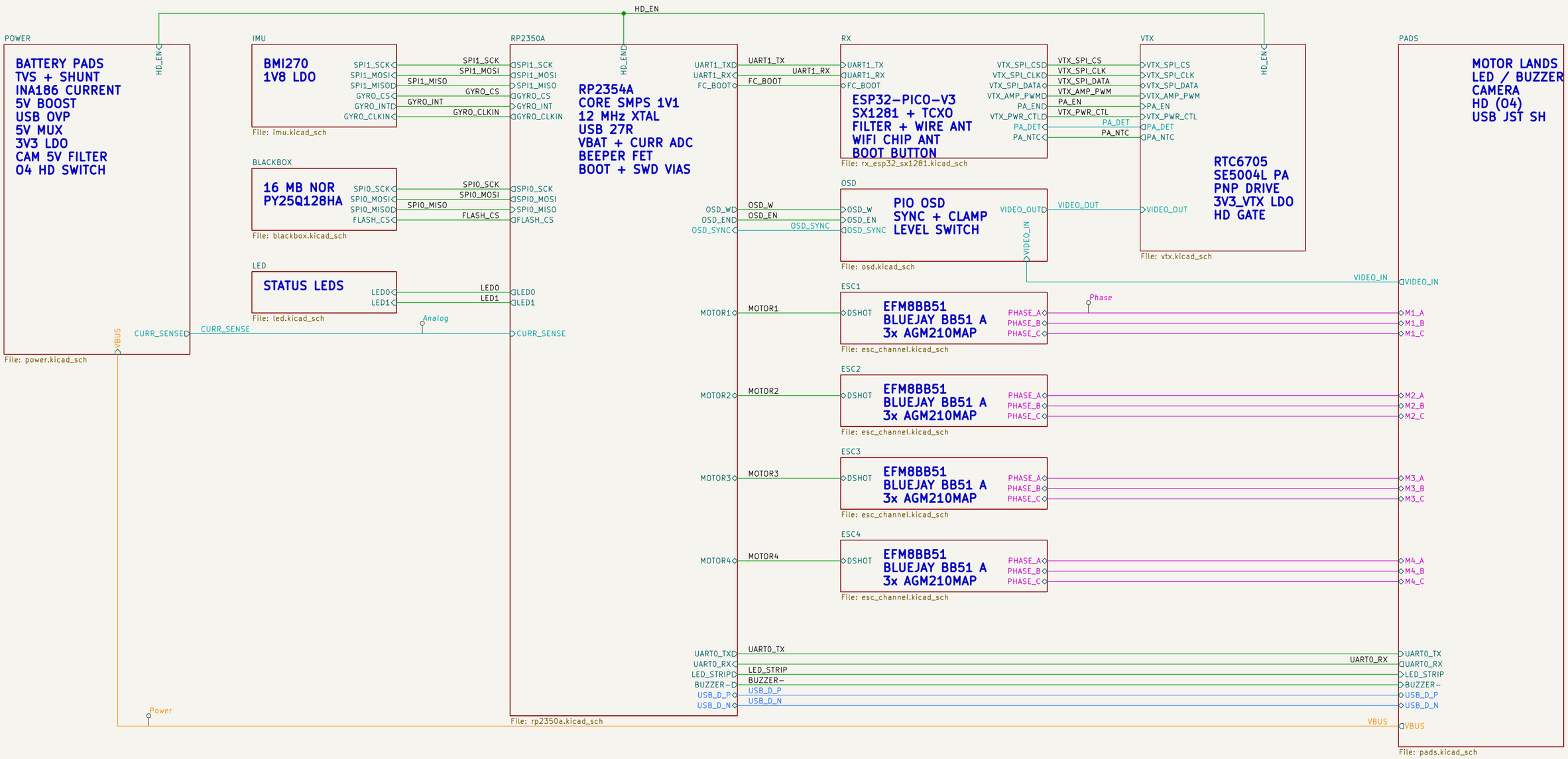


POWER TABLE

Rail	Source	Nominal	Budget (spec 5)	Feeds
CSA+	battery pads J2 / J3 (1S)	3.0–4.35 V	4x ESC + 2.0 A board	TVS, pad bulk, shunt R1
+BATT	shunt R1 0.5 mOhm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, O4 EN, VBAT ADC, PA_VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	U24 OVP
+5V_USB	U24 SGM40661 OVP	5 V	as VBUS	U3 VIN1, PR1 divider
+5V_BST	U2 TPS61022 boost	5.15 V	0.70 A analog / 1.48 A HD (BEC 1.5 A cont.)	U3 VIN2, U5 O4 switch
+5V	U3 TPS2116 mux	5 V	0.70 A analog, about 0.25 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
+5V_HD	U5 TPS22810 (HD_EN, cell)	5 V	1.20 A O4 Lite	VHD pad
+3.3V	U4 LP5912–3.3	3.3 V	0.20 A (0.48 A Wi–Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3.3V_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A typ, about 0.17 A at max PA	RTC6705, PA drive, PA ref
+1.8V_GYRO	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1.1V	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	R67 selector, +BATT default	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
PA_VREF	U22 LP5907–2.85 (EN = PA_EN, off in HD)	2.85 V	about 10 mA	SE5004L VREF/EN (local, VTX)
GND	battery –	0 V	–	all sheets

Generated from hardware/sch_contract.json by hardware/tools/sch_root.py: do not edit by hand. Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE_x). Rails are power symbols (table); VBUS is a hierarchical net. ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

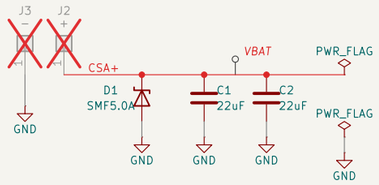
OpenDrone

Sheet: /
File: OpenAIO–Whoop.kicad_sch

Title: OpenAIO–Whoop overview (1S whoop AIO)

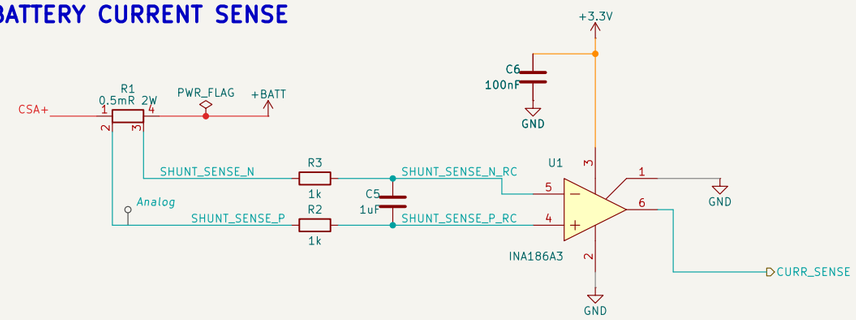
Size: A3	Date: 2026–10–09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 1/14

BATTERY INPUT AND TVS



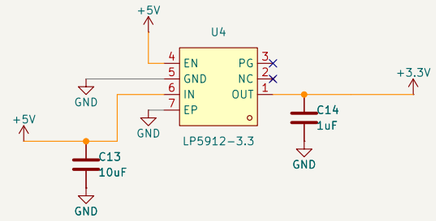
Battery entry from OpenAIO (D113): CSA+ = pad J2 (+) node before the shunt, J3 (−) = GND, 1S 3.0–4.35 V. PWR_FLAGs here because the pads are passive. No reverse-polarity FET (spec 4.1).
D1 SMF5.0A takes one motor's regen pulse on a battery disconnect (fallback SMAJ5.0A); C1/C2 22 uF at the pads. Hot-plug overshoot vs the TPS61022 4.8 V start-up limit is bench item V3c.

BATTERY CURRENT SENSE



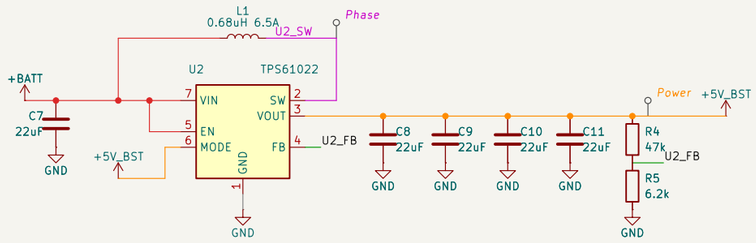
Shunt + INA186A3 from OpenAIO; input RC 0201, common-mode caps deleted (D52). R1 0.5 mOhm Kelvin land: route SHUNT_SENSE_P/N as a pair on L1 or L6, class Analog; every load taps +BATT after R1.
R2/R3 1k + C5 1 uF: 80 Hz. Gain 100 -> 50 mV/A (Betaflight ibata_scale 500), about 65 A full scale. Output RC sits at the ADC pin (RP2350A sheet).

3.3 V LDO FOR FC + RX



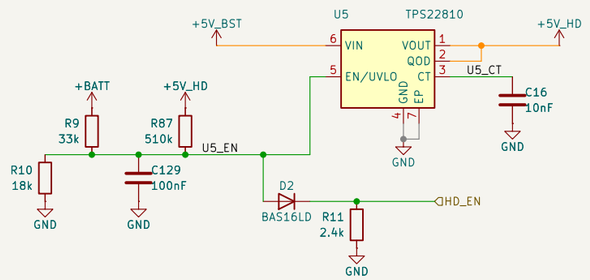
LP5912-3.3 main LDO from OpenFC-Lite-Mini, EN tied to +5V, PG/NC open, EP to GND.
C13 10 uF is the one CIN of U4 and U21 (VTX): place it between the two IN pins. +3.3V COUT sums to about 8.7 uF, inside TI's 0.7–10 uF. 0.2 A load: 0.37 W, +26 K.

5 V BOOST TPS61022



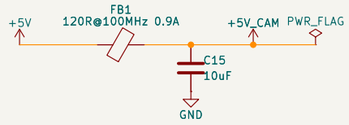
TPS61022 boost (TI SLVSDX7D): VOUT = 0.6 V x (1 + 47k / 6.2k) = 5.15 V; EN on +BATT, MODE = VOUT (forced PWM). C8–C11 4x 22 uF give 20–28 uF effective, so the published BEC is 1.5 A continuous (D79). No feed-forward cap fitted (fit 1.5–2.2 nF across R4 only if V1b rings).
Layout (D92): C8–C11 at pins 3/1, C7 at 7/1, minimal SW copper, R4/R5 at FB away from SW.

O4 LITE HD SUPPLY SWITCH TPS22810



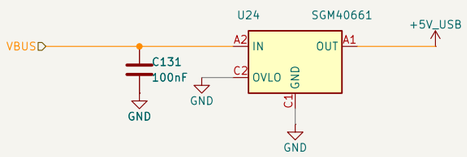
TPS22810 O4 supply switch (D15, D34, D53, D83): VIN on +5V_BST ahead of the mux, so USB never reaches the O4; C8–C11 are its input caps (C17 deleted). C16 10 nF soft start, QOD empties +5V_HD when off.
EN/UVLO = R9/R10 divider from +BATT with R87 hysteresis: on at 3.27–3.77 V, shed at 2.78–3.10 V. HD_EN (GPIO19) low or floating pulls EN below VENF through D2 + R11: O4 off; high: O4 on.

CAMERA 5 V FILTER



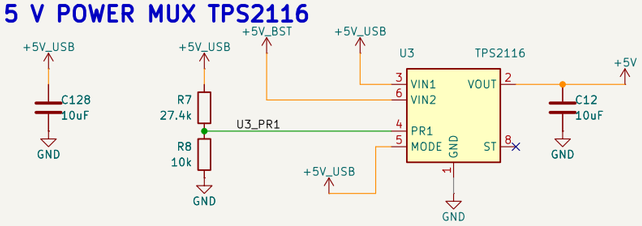
Camera 5 V feed from OpenFC-Lite-Mini: FB1 120 Ohm 0.9 A bead keeps the boost and the camera's switching apart (19 mV at 0.12 A), C15 10 uF after it. +5V_CAM feeds J24 and J32 (PADS); PWR_FLAG because FB1 is passive.

USB INLET OVP SGM40661



SGM40661 USB OVP (D57): IN = VBUS (J31), OUT = +5V_USB. Internal OVLO 5.75–6.12 V: 15 ms debounce + 2 ms soft turn-on keeps a cable ring off the TPS2116 (6 V abs max) and limits the VBUS surge. C131 100 nF at IN, C128 10 uF at U3 VIN1.

5 V POWER MUX TPS2116



TPS2116 mux, priority mode (D24): USB (VIN1) is selected above 3.44–4.04 V, +5V_BST (VIN2) otherwise; both channels block reverse current, so USB never feeds +5V_BST or the O4 (on +5V_BST ahead of the mux, D53). C128 at U3 pin 3, C12 10 uF hold-up at VOUT.

SHEET NOTES

POWER sheet: generated by sch_build.py + sch_root.py, do not edit by hand. Rails are power symbols (net = Value): sheet pins HD_EN in, CURR_SENSE out, VBUS in. R11 is the HD_EN (GPIO19, D111 map) pull-down. Caps stay X5R (D69); one 22 uF hot-loop cap per AGM210MAP on the ESC sheet (D77, D108).
Boost ground (D92): U2 pin 1, C7 and C8–C11 GND form one component-layer island with no via inside the loop; it joins the planes through >= 4 vias at the C7 end, L2 stays solid under it (D85). Keep it away from the shunt traces, gyro, crystals, video ground and antennas. Values: research/DESIGN-SPEC.md 4.1, 4.2, 15.5; history: research/SCHEMATIC.md.

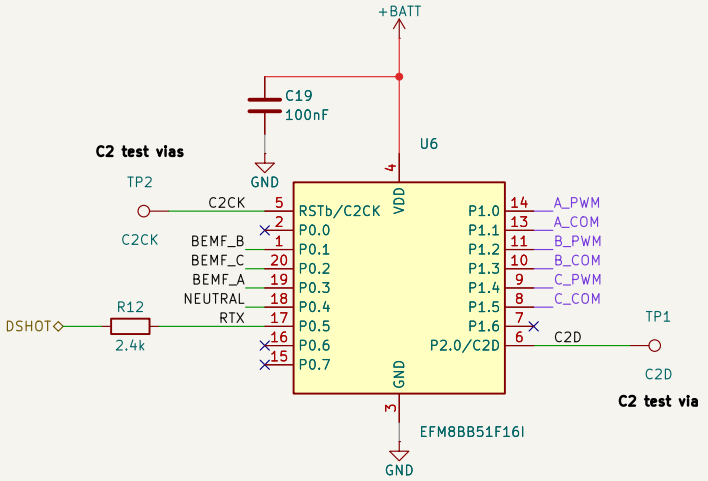
OpenDrone

Sheet: /POWER/
File: power.kicad_sch

Title: Power: input, current sense, 5 V, 3.3 V, O4

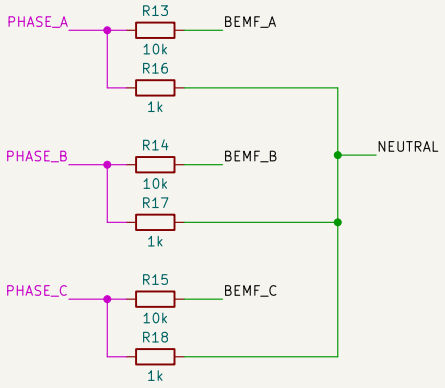
Size: A3	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 2/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pins per Bluejay BB51 layout A: P0.1–P0.3 B/C/A comparator, P0.4 neutral, P0.5 RTX (DShot), P1.0–P1.5 x_PWM (N gate, active high) / x_COM (P gate, active low), P2.0 C2D. Unused pins no-connect.
VDD on +BATT with C19 100 nF at pin 4; the phase B hot-loop cap C23 is its bulk (D108); no ESC3 extra cap (D77). R12 2.4k limits the DShot clamp current at a low cell. C2D/C2CK on test vias TP1/TP2 (D67).

BEMF network: 10k series + 1k neutral star

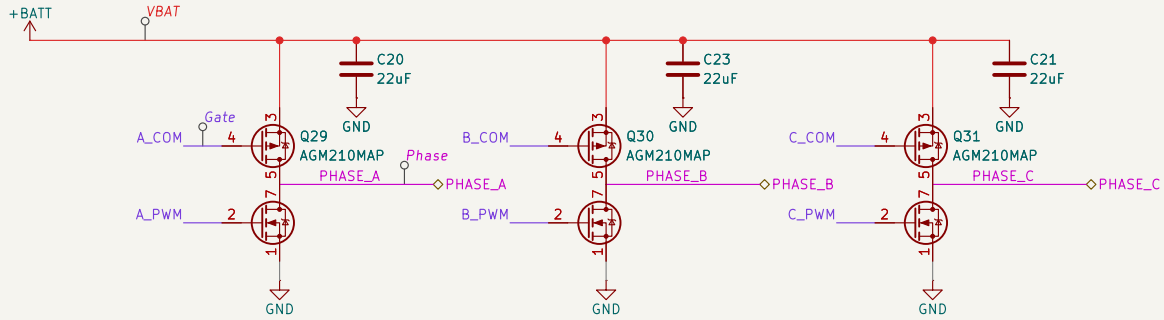


BEMF network from tinyPEPPER: 10k from each phase to its comparator input, 1k from each phase to the virtual neutral (P0.4). No leg to GND at 1S; the 10k keeps the input clamp current < 0.1 mA.

ESC channel, one file placed 4x; refs drawn are ESC1's, the root offsets them per instance:
ESC1 M1 rear right: U6, Q29–Q31, TP1/TP2
ESC2 M2 front right: U7, Q32–Q34, TP3/TP4
ESC3 M3 rear left: U8, Q35–Q37, TP5/TP6
ESC4 M4 front left: U9, Q38–Q40, TP7/TP8
DSHOT in from the FC (GPIO4/6/18/20), PHASE_A/B/C out to the motor lands.
Adapted from OpenAIO esc_channel: EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, D56).

Power-up: the EFM8 runs from the cell before the FC's 3.3 V is up; its weak pull-up can put <= 60 uA on the FC DShot pad (0.72 V over the unpowered-FT limit, current-limited). Accepted on Matrix parity; V9 scopes it. USB only: DShot idle-high back-feeds +BATT through the EFM8 I/O diode (about 1 mA per channel).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N pair per phase (D23 pinout check): S1 = GND, G1 = x_PWM, S2 = +BATT, G2 = x_COM, both drains = PHASE_x (joined only by the phase island: never split it). Gates driven straight from the EFM8, no resistors (Matrix II practice).
Hot loop (D77, D108): one 22 uF 0603 per FET from S2 to S1 at the pins, microvias in both pads. Bluejay dead time A_X_10_96 at first power-up, A_X_5_96 after the V2 shoot-through check.

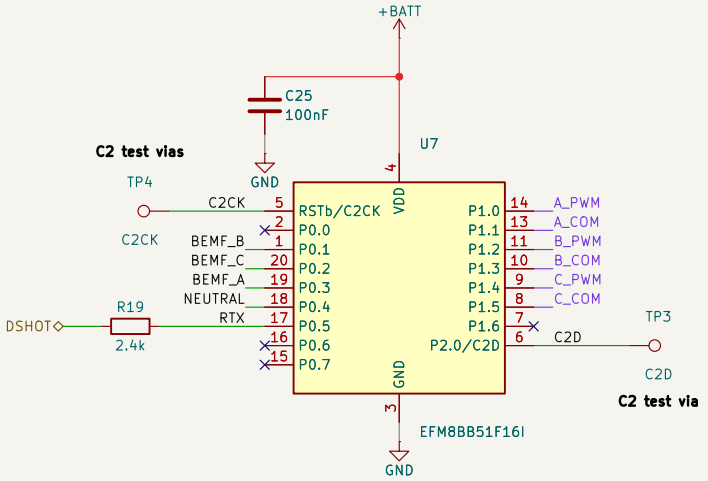
OpenDrone

Sheet: /ESC1/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

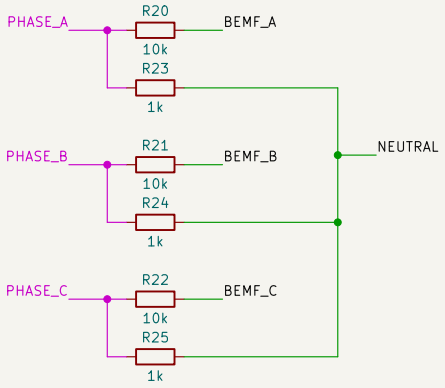
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 3/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pins per Bluejay BB51 layout A: P0.1–P0.3 B/C/A comparator, P0.4 neutral, P0.5 RTX (DShot), P1.0–P1.5 x_PWM (N gate, active high) / x_COM (P gate, active low), P2.0 C2D. Unused pins no-connect.
VDD on +BATT with C19 100 nF at pin 4; the phase B hot-loop cap C23 is its bulk (D108); no ESC3 extra cap (D77). R12 2.4k limits the DShot clamp current at a low cell. C2D/C2CK on test vias TP1/TP2 (D67).

BEMF network: 10k series + 1k neutral star

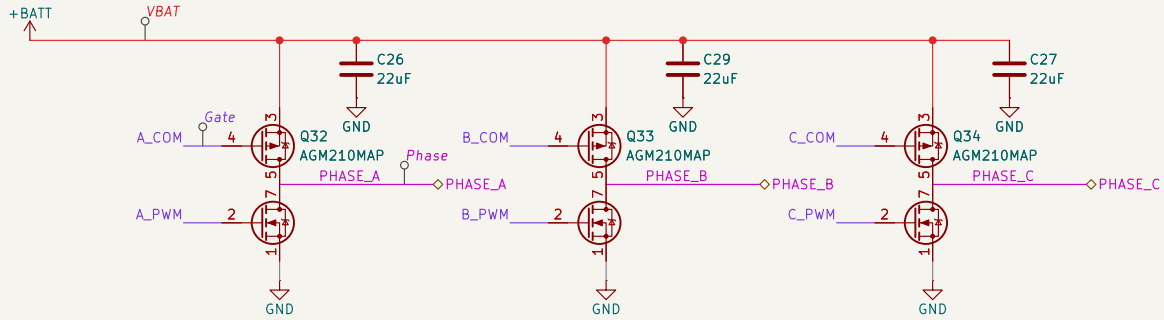


BEMF network from tinyPEPPER: 10k from each phase to its comparator input, 1k from each phase to the virtual neutral (P0.4). No leg to GND at 1S; the 10k keeps the input clamp current < 0.1 mA.

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ESC4 M4 front left: U9, Q38–Q40, TP7/TP8
DSHOT in from the FC (GPIO4/6/18/20), PHASE_A/B/C out to the motor lands. Adapted from OpenAIO esc_channel: EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, D56).

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Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N pair per phase (D23 pinout check): S1 = GND, G1 = x_PWM, S2 = +BATT, G2 = x_COM, both drains = PHASE_x (joined only by the phase island: never split it). Gates driven straight from the EFM8, no resistors (Matrix II practice).
Hot loop (D77, D108): one 22 uF 0603 per FET from S2 to S1 at the pins, microvias in both pads. Bluejay dead time A_X_10_96 at first power-up, A_X_5_96 after the V2 shoot-through check.

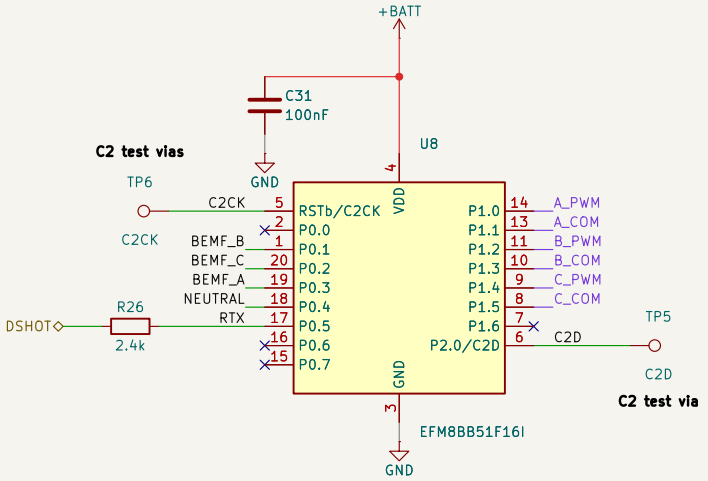
OpenDrone

Sheet: /ESC2/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

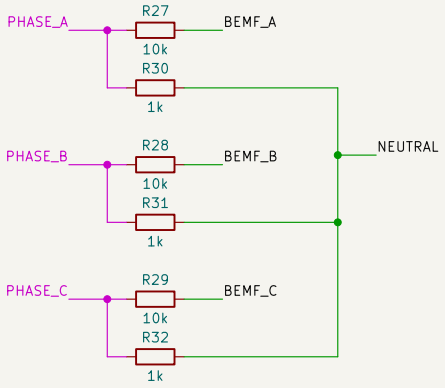
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pins per Bluejay BB51 layout A: P0.1–P0.3 B/C/A comparator, P0.4 neutral, P0.5 RTX (DShot), P1.0–P1.5 x_PWM (N gate, active high) / x_COM (P gate, active low), P2.0 C2D. Unused pins no-connect.
VDD on +BATT with C19 100 nF at pin 4; the phase B hot-loop cap C23 is its bulk (D108); no ESC3 extra cap (D77). R12 2.4k limits the DShot clamp current at a low cell. C2D/C2CK on test vias TP1/TP2 (D67).

BEMF network: 10k series + 1k neutral star

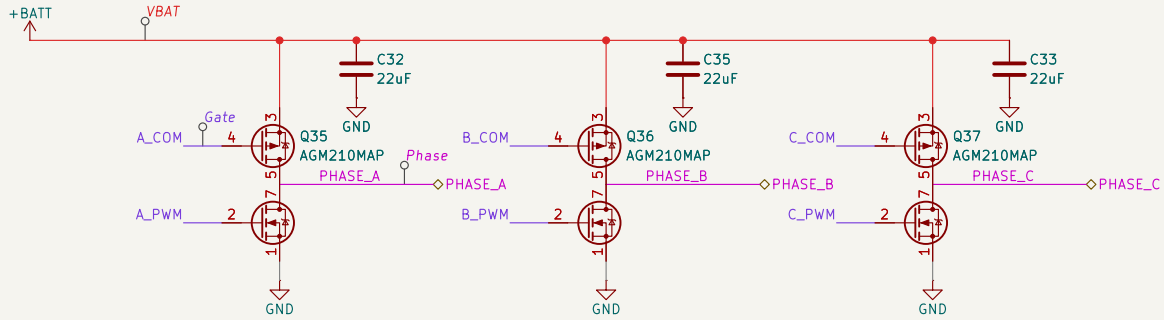


BEMF network from tinyPEPPER: 10k from each phase to its comparator input, 1k from each phase to the virtual neutral (P0.4). No leg to GND at 1S; the 10k keeps the input clamp current < 0.1 mA.

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DSHOT in from the FC (GPIO4/6/18/20), PHASE_A/B/C out to the motor lands. Adapted from OpenAIO esc_channel: EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, D56).

Power-up: the EFM8 runs from the cell before the FC's 3.3 V is up; its weak pull-up can put <= 60 uA on the FC DShot pad (0.72 V over the unpowered-FT limit, current-limited). Accepted on Matrix parity; V9 scopes it. USB only: DShot idle-high back-feeds +BATT through the EFM8 I/O diode (about 1 mA per channel).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N pair per phase (D23 pinout check): S1 = GND, G1 = x_PWM, S2 = +BATT, G2 = x_COM, both drains = PHASE_x (joined only by the phase island: never split it). Gates driven straight from the EFM8, no resistors (Matrix II practice).
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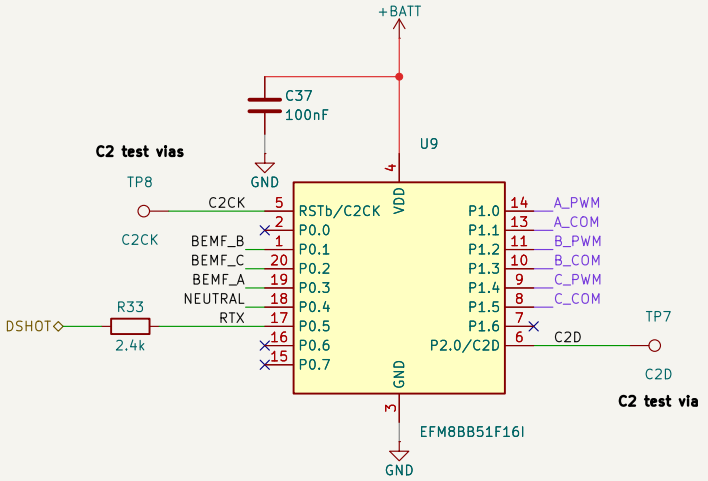
OpenDrone

Sheet: /ESC3/
File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

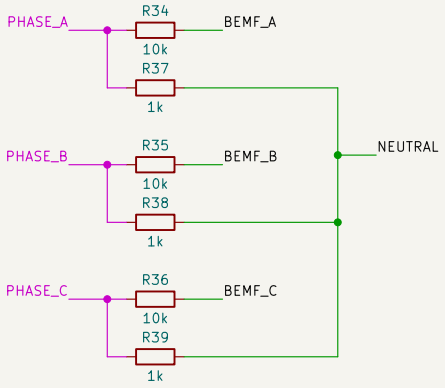
Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 5/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pins per Bluejay BB51 layout A: P0.1–P0.3 B/C/A comparator, P0.4 neutral, P0.5 RTX (DShot), P1.0–P1.5 x_PWM (N gate, active high) / x_COM (P gate, active low), P2.0 C2D. Unused pins no-connect.
VDD on +BATT with C19 100 nF at pin 4; the phase B hot-loop cap C23 is its bulk (D108); no ESC3 extra cap (D77). R12 2.4k limits the DShot clamp current at a low cell. C2D/C2CK on test vias TP1/TP2 (D67).

BEMF network: 10k series + 1k neutral star

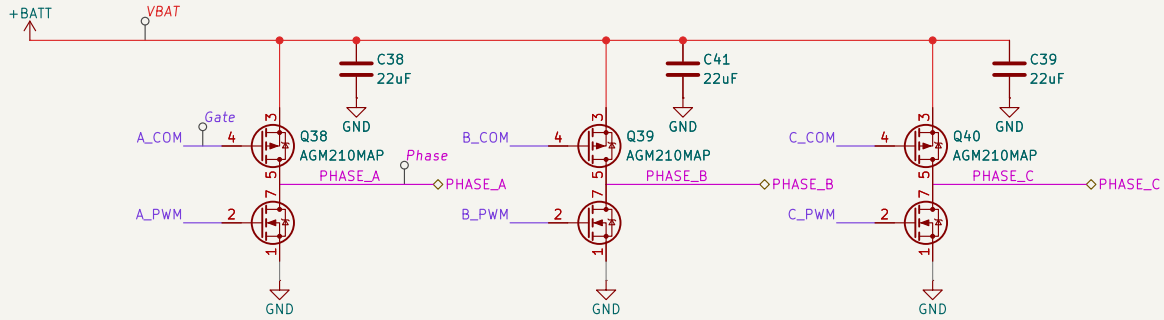


BEMF network from tinyPEPPER: 10k from each phase to its comparator input, 1k from each phase to the virtual neutral (P0.4). No leg to GND at 1S; the 10k keeps the input clamp current < 0.1 mA.

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ESC4 M4 front left: U9, Q38–Q40, TP7/TP8
DSHOT in from the FC (GPIO4/6/18/20), PHASE_A/B/C out to the motor lands. Adapted from OpenAIO esc_channel: EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, D56).

Power-up: the EFM8 runs from the cell before the FC's 3.3 V is up; its weak pull-up can put <= 60 uA on the FC DShot pad (0.72 V over the unpowered-FT limit, current-limited). Accepted on Matrix parity; V9 scopes it. USB only: DShot idle-high back-feeds +BATT through the EFM8 I/O diode (about 1 mA per channel).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP P+N pair per phase (D23 pinout check): S1 = GND, G1 = x_PWM, S2 = +BATT, G2 = x_COM, both drains = PHASE_x (joined only by the phase island: never split it). Gates driven straight from the EFM8, no resistors (Matrix II practice).
Hot loop (D77, D108): one 22 uF 0603 per FET from S2 to S1 at the pins, microvias in both pads. Bluejay dead time A_X_10_96 at first power-up, A_X_5_96 after the V2 shoot-through check.

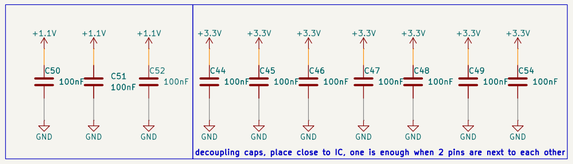
OpenDrone

Sheet: /ESC4/
File: esc_channel.kicad_sch

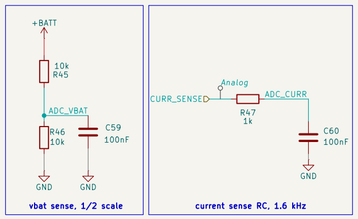
Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 6/14

RP2354A: Smaller variant with integrated flash (2MB)



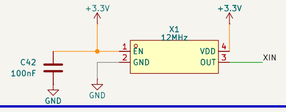
100 nF X6S 0201 per supply pin: DVDD C50–C52, IOVDD C44–C48; C49 at ADC_AVDD 44 (also IOVDD 45), C54 <= 1 mm from QSPI_IOVDD 54 (also USB_OTP_VDD 53).



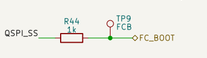
VBAT: R45/R46 10k/10k, 4.35 V -> 2.18 V on GPIO27 (ADC1). vbat_scale 200/100. CURR: R47 1k + C60 100 nF on GPIO26 (ADC0). ibata_scale 500.

12 MHz CMOS oscillator on XIN, XOUT open (D100)

X1 12 MHz CMOS oscillator drives XIN (pin 21); XOUT stays open, no trace (D99, D100 replace the D78 crystal). C42 at X1 VDD; X1 within 3 mm of pin 21.

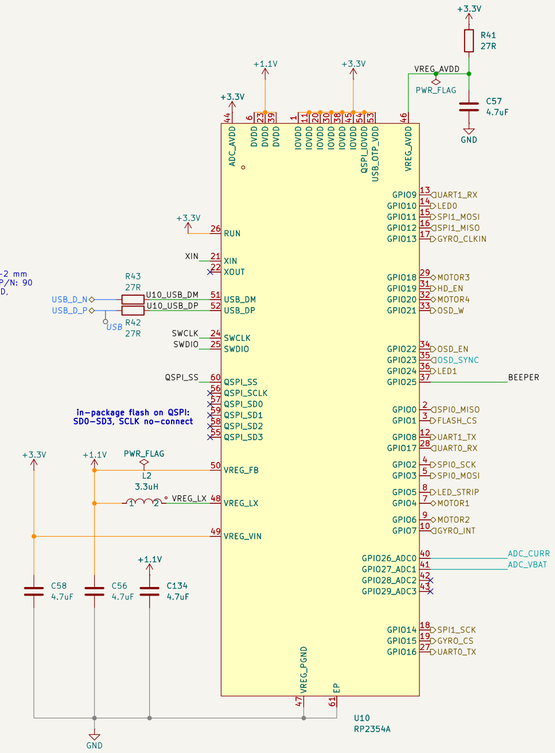


Boot strap + SWD test vias



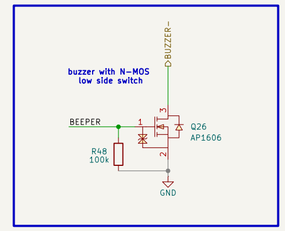
R44 1k QSPI_SS -> FC_BOOT: the shared button SW1 (RX sheet, D65) pulls it low through D8 low at reset = BOOTSEL. Keep the QSPI_SS-to-R44 stub <= 1 mm.
TP9 FCB, TP12 SWCLK, TP13 SWDIO: filled test vias (D67), out of the BOM.





R42/R43 27R within 1–2 mm of pins 52/51. USB_D_P/N: 90 Ohm pair over solid GND, length-matched (D33).

1.1 V SMPS: follow RP2350 DS 6.3.8 placement. C58 CIN, L2, C56 COUT on U10's side within 3 mm of pins 49/48/50/46. VREG_FB from the C56 pad, one-point GND to VREG_PGND, no copper under L2 on the next layer. L2 is polarity-marked (pin 2 = VREG_LX). C134 = 2nd 4.7 uF at DVDD pin 23. R41 + C57 filter VREG_AVDD.



Q26 AP1606 low-side buzzer switch (from OpenFC-Lite-Mini): GPIO25 high = on (BEEPER_INVERTED, active buzzer); R48 100k holds it off through reset. BZ- pad on PADS.

RP2350A sheet, copied block by block from OpenFC-Lite-Mini rp2350a (flown Rev2). Changes: USB on J31 (PADS) with 27R R42/R43 here (D62); no I2C pull-ups, no WS2812 inverter; LEDs on the LED sheet; boot via R44 + FC_BOOT; SWD on test vias (D67); 12 MHz CMOS oscillator (D100); GPIO map D111. Betaflight config: PINMAP 5.

GPIO map (D111, placement V4_B4; research/PINMAP.md):
SPI0 blackbox GPIO0 MISO / 2 SCK / 3 MOSI, FLASH_CS GPIO1.
SPI1 gyro GPIO11 MOSI / 12 MISO / 14 SCK, GYRO_CS 15, GYRO_INT 7, GYRO_CLKIN 13.
UART1 GPIO8/9 = CRSF to the ESP32; UART0 GPIO16/17 = TX1/RX1 pads (O4 MSP, D89).
DShot MOTOR1–4 GPIO4/6/18/20 (PIO0). LED_STRIP 5, LED0 10, LED1 24, BEEPER 25. HD_EN 19 (PINIO1, high = O4 on).
OSD W/EN/SYNC 21/22/23 (PIO2). ADC_CURR 26. ADC_VBAT 27. GPIO28/29 no-connect.

Power-up and 5 V-tolerant pads (PINMAP 4.5, F16): an FT pad takes <= 3.63 V while IOVDD is off.
- MOTOR1–4: the EFM8 pull-ups reach the pads through 2.4k before +3.3V (<= 60 uA, current-limited); Matrix parity, V9 scope.
- HD_EN (GPIO19): R11 2.4k holds it low through reset, so the O4 stays off and the analog VTX on (V9 hot-soak boot check).
- TX1/RX1 as the free UART (D89): 3.3 V peripherals only; a 5 V TX would exceed 3.63 V at power-up.
- GPIO28/29 spare, no-connect (D111 map).

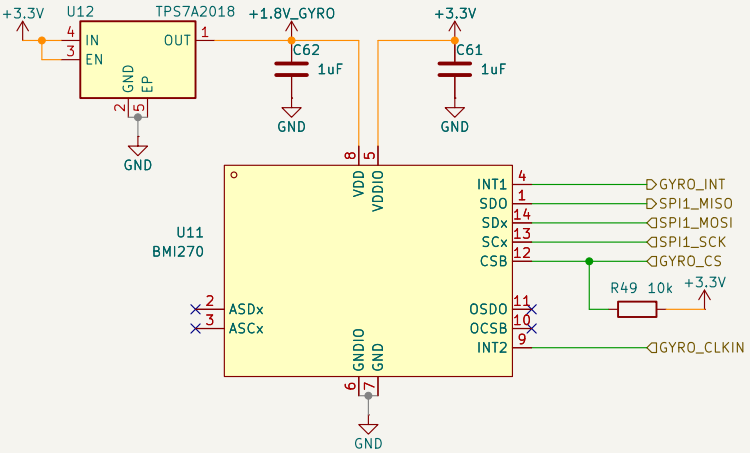
OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

Title: RP2354A flight controller MCU

Size: A2	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 7/14

Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270 (U11) on the universal LGA-14 land from OpenFC-Lite-Mini; ICM-42688-P qualified on the same land (V10 fly-off).
ASDx/ASCx and OCSB/OSDO no-connect (never GND for the BMI270). VDDIO = +3.3V, VDD = +1.8V_GYRO. GYRO_CLKIN (GPIO13) used only with the TDK part.
R49 10k keeps CSB high while GPIO15 resets with its pull-down on. GYRO_1_ALIGN from the placed orientation.

+1.8V_GYRO LDO U12 TPS7A2018 from +3.3V (D52), load < 1 mA. C61 = CIN at gyro VDDIO pin 5, C62 = COUT <= 1.5 mm from VDD pin 8; no separate 100 nF.

OpenDrone

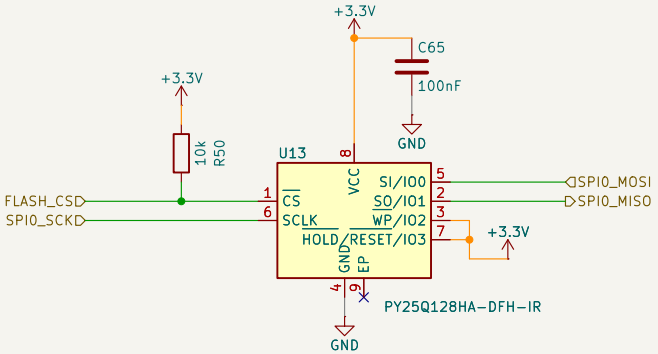
Sheet: /IMU/
File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 8/14

SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)

From OpenFC-Lite-Mini blackbox: microSD replaced by a 16 MB SPI NOR (D66). Betaflight m25p16 driver (JEDEC 85 20 18); SPI0 runs 75 MHz, reads are outside the guaranteed timing: gate F7 / V9 (full log CRC), fallback a lower read clock.
WP# and HOLD#/RESET# tied to +3.3V (pin 7 must not float); the ERC pin_to_pin warning on pins 3/7 is this deliberate tie.
R50 10k keeps CS# high through reset. EP tabs: no-connect, no vias.



Placement: U13, SPI0 and FLASH_CS >= 3 mm from the RX antenna hole (SPI0 harmonics fall in the ELRS band, V7 measures desense). C65 at pin 8.
Sourcing by MPN (D60); same-pinout fallbacks PY25Q128HA-QVH-IR, GD25Q128EQIGR.

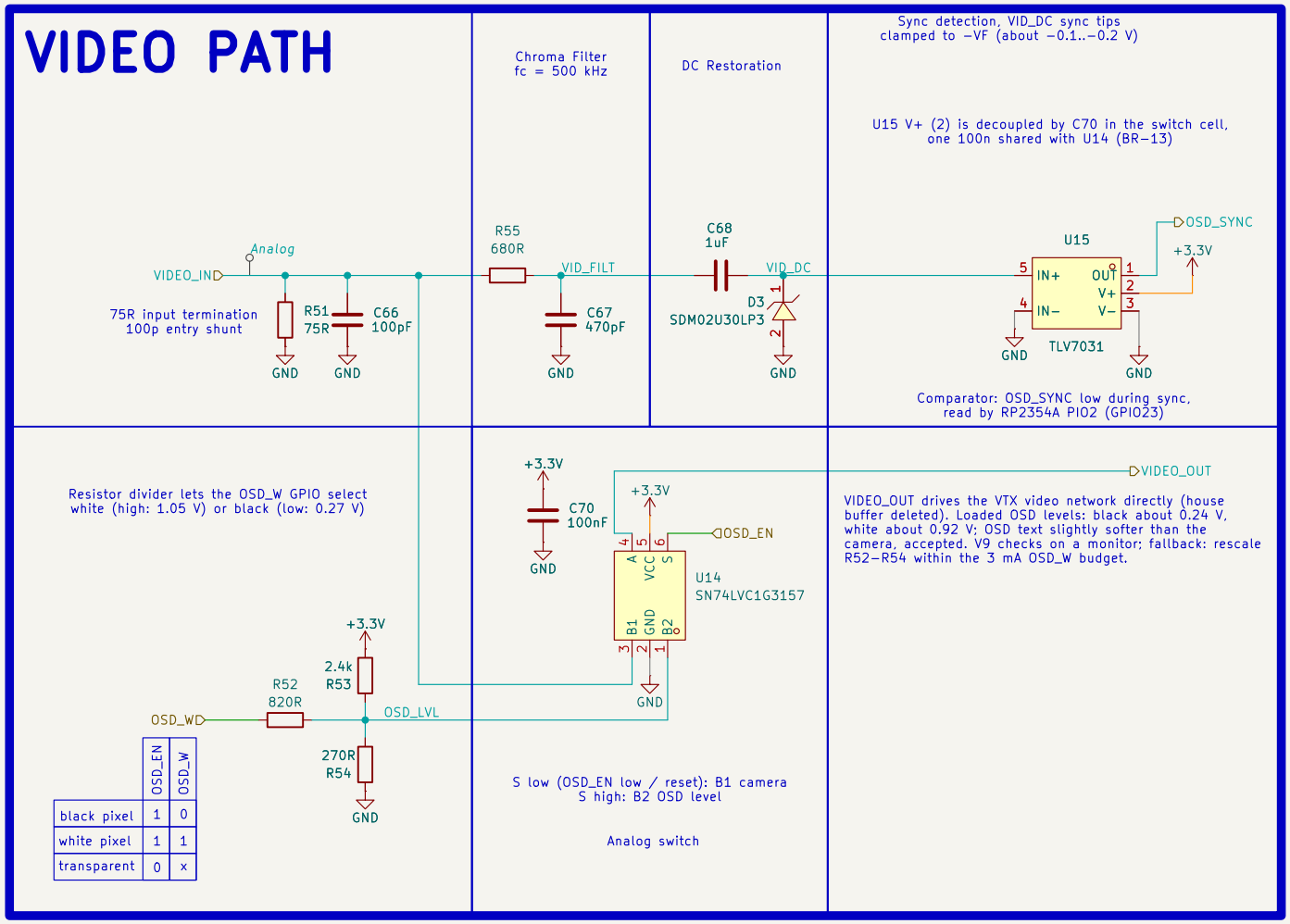
OpenDrone

Sheet: /BLACKBOX/
File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 9/14

VIDEO PATH



Copied from OpenFC−Lite−Mini osd (flown): 75R termination, 680R/470p chroma filter, sync−tip clamp, TLV7031 sync comparator, SN74LVC1G3157 camera/OSD switch. Changed: COS8051 buffer deleted, divider R52−R54 about 5x lower (P4−10), one shared C70 (D52).
Operation: OSD_EN low (reset) −> camera passes through; OSD_EN high −> OSD_W high = white, low = black. No OSD without camera sync. OSD_W/EN/SYNC = GPIO21/22/23 (PIO2).
C68 X5R carries <= 0.6 V DC, no DC−bias loss (D69).

Placement: C70 is the one supply cap of U14 and U15, place it between them. Video entry parts R51/C66 at the CAM pad line; video on L1/L6 over GND, never L3; away from the boost (D69). Analog class on the video nets; OSD_W / OSD_EN Default.

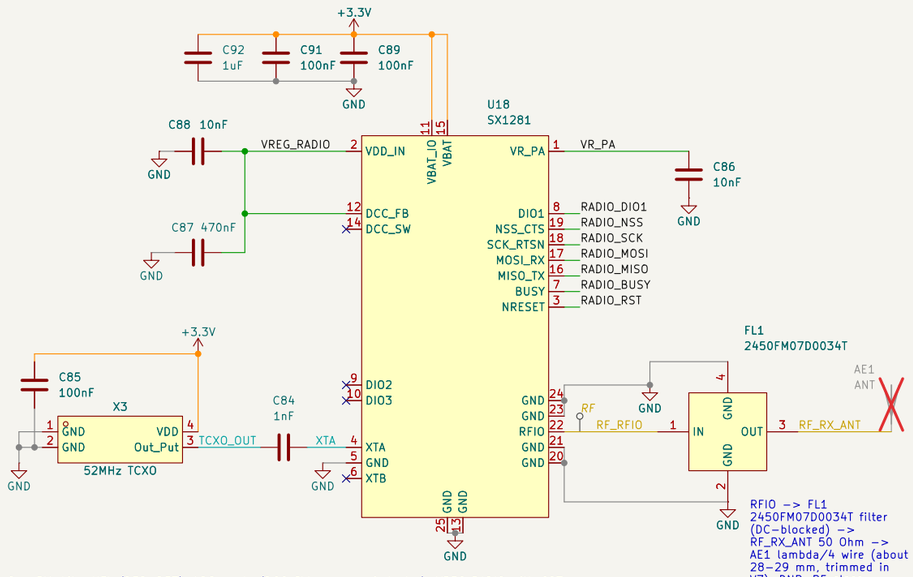
OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

Size: A4	Date: 2026−10−09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 10/14

SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA



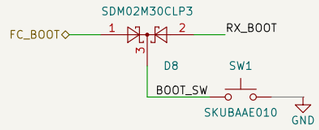
SX1281 from OpenRX-Lite-UFL (D52, D53): LDO mode (DCC_SW open, no inductor), VREG_RADIO with C87 + C88, decoupling per DS Table 15-1. Pin 5 is GND and is grounded here. 52 MHz TCXO X3 AC-coupled by C84 into XTA.

RFIO -> FL1 2450FM07D0034T filter (DC-blocked) -> RF_RX_ANT 50 Ohm -> AE1 lambda/4 wire (about 28-29 mm, trimmed in V7), DNP. RF class.

NOTES

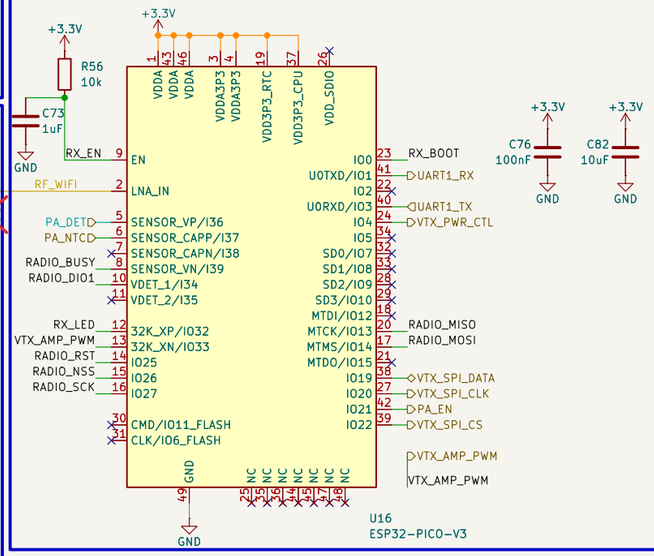
RX sheet: radio, TCXO, RGB LED and Wi-Fi match copied from OpenRX-Lite-UFL (D52, D53, D58); ESP32-PICO-V3 block new from Espressif Fig. 11 (D55); shared boot / bind button new (D65). Pin plan research/PINMAP.md 6, parts hardware/bom_plan.json, +3.3V from POWER.

BOOT / BIND BUTTON (D65)



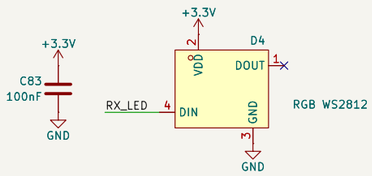
Shared boot / bind button (D65): SW1 to GND, D8 dual Schottky to FC_BOOT (RP2354A QSPI_SS via R44) and RX_BOOT (ESP32 GPIO0). Released, the diodes isolate the two straps.
Held at power-up: RP2354A BOOTSEL + ESP32 download mode (recovery, D67). Pressed while running: ELRS bind / Wi-Fi / reboot, the FC keeps running.

ESP32-PICO-V3 MCU (ELRS)



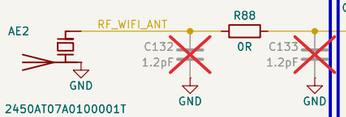
ESP32-PICO-V3 (D55): all supplies on +3.3V, C82 + C76 at VDDA3P3; flash, crystal and SDIO cap are inside the SiP. EN = RX_EN with R56 10k + C73 1 uF.
Straps (D111 map): GPIO0 = RX_BOOT; GPIO2, 5, 12, 15 not connected (internal defaults). PA_EN on GPIO21, PA_DET on GPIO36 (ADC1), PA_NTC on GPIO37.
ELRS layout JSON (D53, D81): see research/PINMAP.md 6; vtx_miso must be written as -1.
FIRMWARE RULE: GPIO4 VTX_PWR_CTL must be driven open-drain (low = VTX off, released = on); R68 10k on the VTX sheet keeps stock ELRS running the VTX.
Wi-Fi TX peaks 370 mA (bench mode only); +3.3V budget on POWER. SiP site per floorplan (D61), no halo rule (D69).

ELRS RGB LED



D4 RGB LED (ELRS led_rgb GPIO32) from OpenRX-Lite-UFL. P4-15 DECIDED: keep on +3.3V as the house RX boards do (0.2 V under the datasheet VDD minimum; on +5V the DIN level would be short). Bench V-LED: fallback VDD to +5V.

Wi-Fi CHIP ANTENNA (D58)

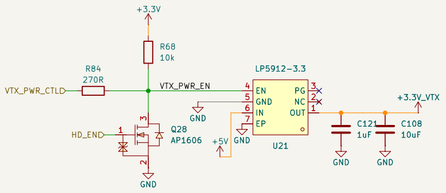


Wi-Fi (D58): LNA_IN -> RF_WIFI 50 Ohm (<= 8 mm, D69) -> pi match C133 (DNP) / R88 0R / C132 (DNP) -> AE2 chip antenna; values from V7 tuning. Keep-out per DRU (D21).

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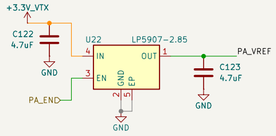
Sheet: /RX/		
File: rx_esp32_sx1281.kicad_sch		
Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281		
Size: A3	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 11/14

+3.3V_VTX LDO + HD GATE



U21 LP5912-3.3 VTX supply on +5V: CIN = shared C13 on POWER if within 1 cm, else its own 10 uF.
EN = VTX_PWR_EN: R68 10k to +3.3V keeps the VTX on with stock ELRS (GPIO4 has a weak reset pull-down).
FIRMWARE RULE: patched ELRS drives ESP32 GPIO4 OPEN-DRAIN through R84 270R: low = VTX off, released = on. Never drive GPIO4 push-pull high (12 mA into Q28 in HD mode, fault case).
Q28 AP1606 on HD_EN pulls EN low in HD mode (VTX off). +3.3V_VTX COUT about 6.8 uF effective; C108 at the RTC6705 VCO pins 39/40.

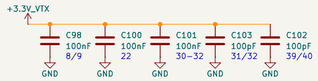
PA REFERENCE LP5907-2.85



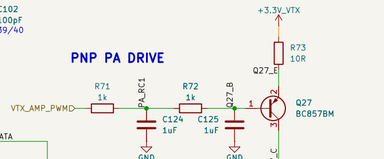
U22 LP5907-2.85 PA reference from +3.3V_VTX: EN = PA_EN (ESP32 GPIO21, D111 map); its 1 MOhm EN pull-down holds the PA off until ELRS drives it. C122 4.7 uF CIN within 1 cm, C123 4.7 uF out.
Bench-gated: the VREF window vs the SE5004L (V5) and the HD switch-on excursion of PA_VREF (V9): fallback C123 1 uF or patched ELRS drops PA_EN on HD_EN.

RTC6705 VTX: 8 MHz CRYSTAL, SPI, VIDEO, PLL, PA DRIVE, PAOUT1 FEED

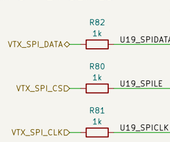
Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):



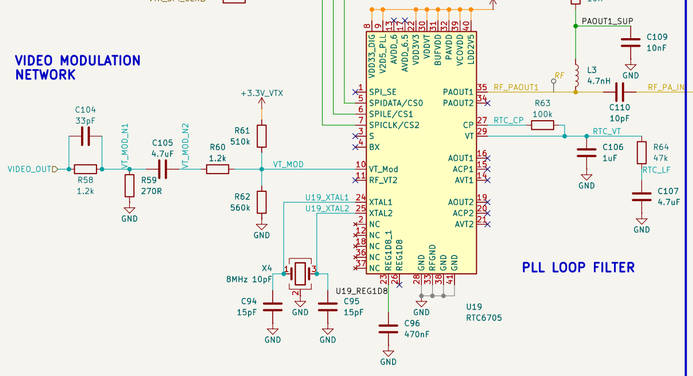
PNP PA DRIVE



VTX SPI 1k SERIES



VIDEO MODULATION NETWORK



PLL LOOP FILTER

RTC6705 checked pin by pin against OpenOSD-X v1.01: decoupling per pin as drawn (numbers next to each cap): REG1D8, AVDD_6.5 and RF_VT2 open (V5 spur checks). X4 8 MHz exactly (ELRS R = 400), CL 10 pF.

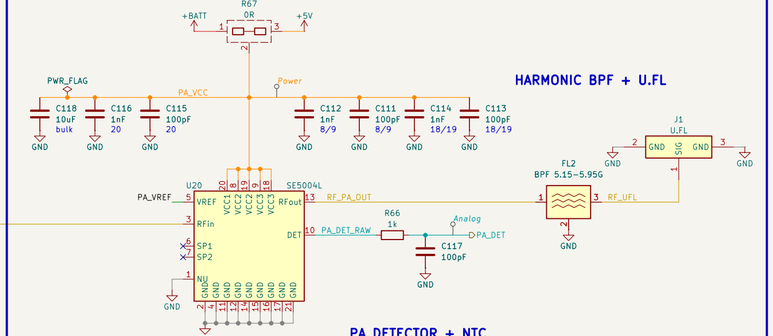
VTX SPI: R80/R81/R82 1k in series, PLACED AT THE ESP32 END: they limit back-feed into the unpowered RTC6705 in HD mode. 1 MHz, the RC is far below the timing limits; SPIDATA is bidirectional (3-wire reads).

Video modulation and PLL loop filter as OpenOSD-X v1.01: VIDEO_OUT -> R58 || C104, R59, C105, R60 -> VT_MOD with R61/R62 bias. Loop filter CP -> R63 -> VT, C106, R64 + C107. RTC_CP / RTC_VT / RTC_LF / VT_MOD are high-impedance (Analog class): short copper, >= 5 mm from the boost and ESC phases.

PNP PA drive: ESP32 GPIO33 VTX_AMP_PWM (ELRS vtx_amp_pwm, D111 map) -> 2 RC poles -> Q27 BC857 with emitter R73 -> PAOUT1 supply: sets the RTC6705 output-stage current (not the SE5004L bias). GPIO33 is high-impedance from reset and the PA stays off (PA_EN), so boot leakage is bounded (V5). Start value R73 = 10R; at the top step R73/R65/Q27 run near their derated ratings: V8 measures, levers 27R or a capped ELRS power table.

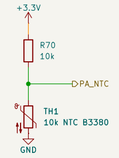
PAOUT1 feed: L3 4.7 nH choke from PAOUT1_SUP (C109), C110 10 pF DC block into the PA input ('DC block required'). Only L3, C109, C110 between pin 35 and the PA: 50 Ohm CPWG < 5 mm, RF class.

SE5004L PA + SUPPLY, DETECTOR, BPF + U.FL



HARMONIC BPF + U.FL

PA DETECTOR + NTC



SE5004L PA (EK1 layout): RFout has an internal DC shunt, so RFout -> FL2 directly, no match parts. PA_VCC = R67 0R selector: +BATT (default, pads 1-2) or +5V (pads 2-3, rework). EK1 decoupling at 8/9, 18/19 and 20 + C118 10 uF. EP and GND pins to a via field.

PA_DET: DET -> R66 1k / C117 100p -> ESP32 GPIO36 (ADC1): keep the RC at the ESP32 end. PA_NTC: TH1 in the PA ring + R70 10k -> GPIO37 (1.65 V at 25 C). FL2 BPF on the PA side; J1 U.FL with L2 cut out under the signal pad, >= 4 GND vias.

SHEET NOTES

VTX sheet, generated by gen_vtx.py. No sibling exists: circuit per the OpenOSD-X v1.01 reference, the SE5004L datasheet + EK1 and DESIGN-SPEC 4.8. Flow: ESP32 controls (RX sheet) left, RTC6705 middle, PA, BPF and U.FL right.

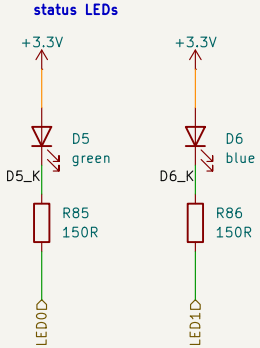
BENCH ITEMS: RF_VT2 spur (V5): PA drive Re (V5/V8): PA_VREF window and HD switch-on (V5/V9); OSD contrast (V9): X4 start-up margin (V5): top-step temperature of Q27/R73/R65 (V5/V8).

OpenDrone

Sheet: /VTX/
File: vtx.kicad_sch

Title: 5.8 GHz VTX: RTC6705 + SE5004L PA

Size: A2	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 12/14



Status LEDs from OpenFC-Lite-Mini: +3.3V -> LED -> 150R -> GPIO, the pin sinks (low = on, Betaflight default). LED0 = GPIO10 green D5, LED1 = GPIO24 blue D6.
About 2-3 mA, ceiling 6 mA: LED, resistor and GPIO well inside ratings. The LEDs may glow faintly from the reset pull-downs until Betaflight drives the pins: cosmetic, accepted. Slow signal, R at either end (D33).

OpenDrone

Sheet: /LED/
File: led.kicad_sch

Title: Status LEDs

Size: A5	Date: 2026-10-09	**Rev: rev1**
KiCad E.D.A. 10.0.6		Id: 13/14

Battery pads (1S)

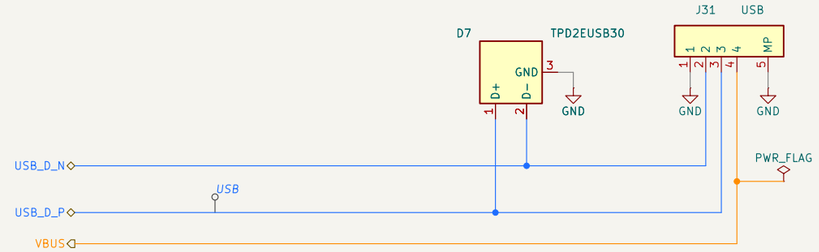
Battery pads J2 (+, CSA+) / J3 (−, GND) are drawn on the POWER sheet (D113): D1.7 mm plated pads (D101) for a user-fitted BT2.0 pigtail, >= 12 power vias each (D72). 1S only, 3.0–4.35 V (D14). Copper only (DNP, A16).

Motor lands M1–M4: plug-ready, Molex PicoBlade header DNP



Motor lands (D65, Matrix 1S 5IN1 II style): one PicoBlade land per motor, pads 1/2/3 = PHASE_C/B/A; phase order is free, Bluejay sets the direction. On the top side, mouths facing out (D72).
Solder the motor wires (default) or fit the DNP Molex 53047-0310 header: its 1.0 A contact rating is below the motor current, plugged motors run at the user's risk (README); V4 records the header temperature.
M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad-X).

USB: JST SH 1.0 mm 4-pin + ESD



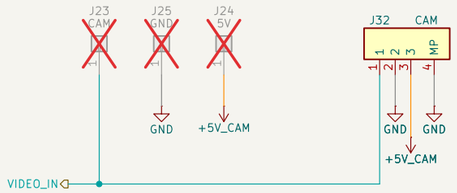
USB: J31 JST SH 4-pin vertical (D76; D72 side-entry withdrawn), BetaFPV order 1 GND, 2 D−, 3 D+, 4 VBUS. Pin 1 = GND is inferred: gate V0-USB before the P5 freeze (a mirrored plug puts −5 V on the OVP input).
D7 TPD2EUSB30 ESD at J31, ahead of the 27R at the MCU. USB pair 90 Ohm, length-matched (D33). VBUS -> POWER (U24 OVP, D57); PWR_FLAG here.

LED strip + buzzer row



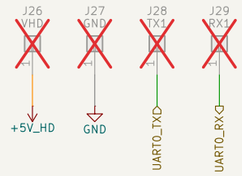
Row GND LED 5V BZ− (D63, D64; own GND pad J33 supersedes D27). LED = LED_STRIP straight from GPIO5, no translator (D52); a dead first LED is the level-problem symptom (V9). 5V feeds the strip and the buzzer, BZ− = drain of Q26 (RP2350A sheet). Copper only (DNP, A16).

Camera pads + JST SUR 0.8 mm plug



Camera: JST SUR 0.8 mm plug J32 and pads CAM / GND / 5V in parallel (D63, D64), one mating zone (D72). VIDEO_IN -> OSD sheet (Analog class): +5V_CAM via FB1 on POWER, 0.12 A.
Gate G1 open: plug pin order provisional until a C03 lead is measured (a mirrored footprint puts 5 V on the camera's video output).

HD pads (DJI O4 Lite)



HD pads for the DJI O4 Lite (D13, D16): VHD = +5V_HD (U5, on with HD_EN), TX1/RX1 = UART0 GPIO16/17 (silk per D89) for MSP DisplayPort; no SBUS pad.
Analog builds: TX1/RX1 are the only free UART (D89). 3.3 V logic only; a 5 V TX on RX1 would exceed the unpowered pad limit during the ramp. Copper only (DNP, A16).

Mounting holes



H1–H3: 25.5 mm M1.4 pattern (D73) cut in Edge.Cuts; symbols excluded from board, BOM and CPL, no net.

PADS: every off-board connection, style from the OpenFC-Lite(-Mini) pads sheets. Solder pads are DNP and out of BOM / CPL (LINEUP A16); the motor header is a DNP BOM line. 20 hierarchical labels = the sheet pins. TX1/RX1 = UART0, the only free UART (D89).
Owed: J31 pin 1 = GND (V0-USB), J32 pin order (G1), motor-header temperature (V4).

OpenDrone

Sheet: /PADS/
File: pads.kicad_sch

Title: Pads, motor lands, camera, HD, USB

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